

Question:

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy .

Reasoning + Answer:

<think>

1. We are given that $\log_x(y^x)=10$ and $\log_y(x^{4y})=10$. We need to find xy .

2. Using the properties of logarithms, we can rewrite the given equations:

$$\log_x(y^x)=x\log_x(y)=10$$

$$\log_y(x^{4y})=4y\log_y(x)=10$$

3. Let us introduce a new variable $u=\log_x(y)$ and $v=\log_y(x)$. Using the properties of logarithms, we can express $\log_y(x)$ in terms of $\log_x(y)$:

$$v=\frac{1}{u}$$

4. Substitute u and v into the given equations:

$$xu=10\quad\text{(1)}$$

$$4yv=10\quad\text{(2)}$$

5. We know that $\log_x(y)=u$ and $\log_y(x)=\frac{1}{u}$, thus:

$$x\cdot u=10\quad\text{and}\quad 4y\cdot\frac{1}{u}=10$$

6. From $xu=10$, we solve for x :

$$x=\frac{10}{u}$$

7. From $4y\cdot\frac{1}{u}=10$, we solve for y :

$$4y\cdot\frac{1}{u}=10\implies y=\frac{10u}{4}=\frac{5u}{2}$$

8. Substitute x and y into xy :

$$xy=\left(\frac{10}{u}\right)\left(\frac{5u}{2}\right)$$

9. Simplify the expression:

$$xy=\left(\frac{10}{u}\right)\left(\frac{5u}{2}\right)=\left(\frac{50u}{2u}\right)=25$$

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<answer>25</answer>